```
#include <iostream>
using namespace std;

char board[3][3] = {
  {'1', '2', '3'},
  {'4', '5', '6'},
  {'7', '8', '9'}
};

void displayBoard() {
  cout << "\n";
  cout << " " << board[0][0] << " | " << board[0][1] << " | " << board[0][2] << endl;
  cout << "---|---|---" << endl;
  cout << " " << board[1][0] << " | " << board[1][1] << " | " << board[1][2] << endl;
  cout << "---|---|---" << endl;
  cout << " " << board[2][0] << " | " << board[2][1] << " | " << board[2][2] << endl;
  cout << "\n";
}

bool checkWin() {
  for (int i = 0; i < 3; i++) {
    if (board[i][0] == board[i][1] && board[i][1] == board[i][2])
      return true;

    if (board[0][i] == board[1][i] && board[1][i] == board[2][i])
```

```
            return true;
        }

        if (board[0][0] == board[1][1] && board[1][1] == board[2][2])
            return true;

        if (board[0][2] == board[1][1] && board[1][1] == board[2][0])
            return true;

        return false;
    }

    int main() {
        int choice;
        int moves = 0;
        char player = 'X';

        cout << "========================\n";
        cout << "     TIC TAC TOE\n";
        cout << "========================\n";

        while (moves < 9) {
            displayBoard();

            cout << "Player " << player << ", enter position (1-9): ";
            cin >> choice;
```

```
if (choice < 1 || choice > 9) {
   cout << "Invalid position! Try again.\n";
   continue;
}

int row = (choice - 1) / 3;
int col = (choice - 1) % 3;

if (board[row][col] == 'X' || board[row][col] == 'O') {
   cout << "Position already occupied! Try again.\n";
   continue;
}

board[row][col] = player;
moves++;

if (checkWin()) {
   displayBoard();
   cout << "Congratulations! Player " << player << " wins!\n";
   return 0;
}

if (player == 'X')
   player = 'O';
else
```

```
            player = 'X';
        }

        displayBoard();
        cout << "Game Draw!\n";

        return 0;
    }
```